

4.2 Creating a SQLite Database in Python

ایس کیو لائٹ پائٹھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔

CS XII

Chapter 4: Data and Analysis

Lecture 03 of 8

4.2 Creating a SQLite Database in Python

LEARN IT

SQLite is built into Python. The whole database is one `.db` file next to your script — no password, no port, no server. Nine steps you must be able to write from memory:

```
import sqlite3
conn = sqlite3.connect("school.db") # 1-2 driver + connection
cur = conn.cursor()                # 3 cursor
cur.execute("CREATE TABLE IF NOT EXISTS students (id INTEGER, name TEXT, marks
INTEGER)")
cur.execute("INSERT INTO students VALUES (1, 'Ali', 85)")
conn.commit()                      # 6 SAVE or the row vanishes
cur.execute("SELECT * FROM students")
rows = cur.fetchall()
for r in rows:
    print(r)
conn.close()                       # 9 unlock the file
```

OUTPUT

```
(1, 'Ali', 85)
```

Step	Why it matters
<code>connect</code>	Creates the file if it does not exist
<code>cursor</code>	Vehicle that carries SQL
<code>execute</code>	Runs one SQL command
<code>commit</code>	Writes INSERT/UPDATE/DELETE to disk
<code>fetchall</code>	Brings SELECT results into a Python list of tuples
<code>close</code>	Frees memory and unlocks the file

COMMON MISTAKE

Forgetting `commit()` is the classic bug: the program prints nothing next time because the insert was never saved. Forgetting `close()` can lock `school.db` so the next run fails.

REMEMBER IT

Strings inside SQL use single quotes: 'Ali'. Python strings around the SQL use double quotes (or a triple-quoted string).

اردو میں سمجھیں

ایس کیو لائٹ پائٹھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔ ترتیب: امپورٹ، کنکٹ، کرسر،
COMMIT, EXECUTE, COMMIT, SELECT, FETCH, CLOSE نہ ہو تو اندراج ضائع ہو جاتا ہے۔

CHECK YOURSELF

1. What does * mean in SELECT *?
2. Which call must follow INSERT?

Answers: 1. All columns. 2. conn.commit().

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. `sqlite3.connect("school.db")` will:

- a. Open or create the file
- b. Always error if missing
- c. Start MySQL server
- d. Draw a pie

MCQ 2. After INSERT you must:

- a. `commit()`
- b. only close
- c. drop the table
- d. `pip install sqlite`

MCQ 3. `fetchall()` returns:

- a. A list of tuples (rows)
- b. A DataFrame always
- c. A dict of dicts always
- d. A string of SQL

MCQ 4. `CREATE TABLE IF NOT EXISTS` avoids:

- a. Error on re-run
- b. The need for types
- c. The need for keys
- d. `commit`

MCQ 5. `close()` is needed to:

- a. Unlock the file
- b. Create pandas
- c. Sort rows
- d. Encrypt OSI

MCQ 6. `SELECT *` means:

- a. All columns
- b. All tables in the folder
- c. A pie of all apps
- d. Drop everything

Extra short questions

1. Nine-step sequence from the notes.
2. What does `*` mean?
3. Two classic bugs.
4. How do you see the data next run?

Extra long question

1. Write a complete program: table `students(id, name, marks)`, insert Ali 85 and Sara 90, print rows. Number the five connection components in comments.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. import, connect, cursor, create, insert, commit, select, fetch, close.
2. Every column.
3. No commit; forgotten close; SQL string quotes mixed up.
4. It is in school.db — connect again and SELECT.

Long

1. Full working code and output two tuples. Comments:
driver/connection/cursor/SQL/database.

Next lecture: 4.3 Basic Operations on Data using Pandas